

A Real-Time Proxy for the Global Financial Cycle

Iván Weigandi
University of Leeds

June 19, 2026

Abstract

This note details the construction of a real-time, open-source proxy for the Global Financial Cycle (GFC) measure of [Miranda-Agrippino and Rey \[2020\]](#). Following [Habib and Venditti \[2019\]](#), I employ a dynamic panel of 57 national and sectoral stock market indices recovered from the archived Stooq database plus documented Yahoo Finance backfills and extract a common factor that captures the co-movement in global financial conditions. The proxy is updated weekly and provides a timely indicator for researchers and practitioners.

1 Motivation

The Global Financial Cycle (GFC) describes the common co-movement of gross capital flows, asset prices, and credit growth across countries. As documented by [Miranda-Agrippino and Rey \[2020\]](#), a significant portion of this variation can be captured by a single global factor. This project provides a transparent, open-source, and automatically updated proxy for this factor using publicly available data, offering a real-time view of global financial conditions based on the methodology of [Habib and Venditti \[2019\]](#)¹.

2 Data

The proxy is constructed from a 57-index historical input panel. The recovered Stooq archive contains 54 of the 57 indices in the original index list; the Dow Jones Composite, Dow Jones Utilities, and Dow Jones Transportation series are backfilled from Yahoo Finance because they are not present in the recovered archive. For current updates, the replication script requires this historical input panel, checks overlapping values between the historical and replacement-source price series by index, splices the country/index price series where a replacement source is available, and then recomputes the

¹See also [Davis and Zlate \[2023\]](#), [Aldasoro, Hördahl & Zhu \[2022\]](#).

GFC proxy from the merged panel. Replacement observations are obtained from Yahoo Finance and public exchange APIs. The full dataset and replication code are available at <https://github.com/iweigandi/daily-global-financial-cycle-proxy>.

3 Methodology

The Global Financial Cycle (GFC) proxy is constructed as the first principal component (PC1) from a panel of national and sectoral stock market index returns. In this section I provide the rationale for working with national and sector-level indices, following the intuition in [Habib and Venditti \[2019\]](#).

3.1 Model Specification

The benchmark framework is the dynamic factor model of [Miranda-Agrippino and Rey \[2020\]](#), which is estimated on a large cross-section of individual asset returns. In their formulation, the return on asset i in country j at time t is decomposed into a global factor, a country-specific factor, and an idiosyncratic shock:

$$r_{i,j,t} = \lambda_i f_t^{\text{global}} + \lambda_{i,j} f_{j,t} + \xi_{i,j,t}. \quad (1)$$

By contrast, the approach of [Habib and Venditti \[2019\]](#) works with country-level averages of returns, rather than individual assets. This leads to a simplified specification:

$$r_{j,t} = \lambda_j f_t^{\text{global}} + \xi_{j,t}. \quad (2)$$

Equation (2) shows that country-level returns load directly on the same global factor f_t^{global} as in the full model, but without the additional country factor $f_{j,t}$. As argued by [Habib and Venditti \[2019\]](#), regional or national averages provide a consistent estimate of the global factor, since they aggregate idiosyncratic components while preserving the common exposure to f_t^{global} . Thus, modeling national indices rather than thousands of individual assets is a valid and tractable alternative for extracting the GFC.

3.2 Proxy Construction

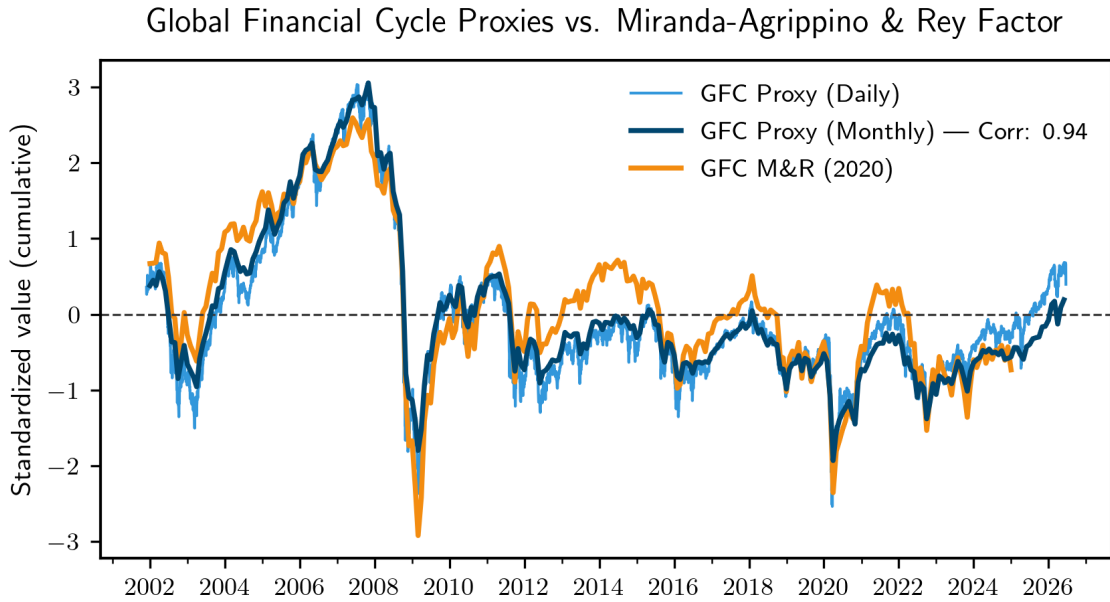
Two versions of the proxy are calculated:

- **Monthly Proxy:** The PC1 is extracted from the balanced panel of monthly log-returns.
- **Daily Proxy:** PC1 is extracted from the balanced panel of unsmoothed daily log-returns.

4 Results

The current construction uses the recovered 57-index historical input panel as the canonical universe. Replacement-source prices are used only to extend those same index series; replacement-only series are excluded from the PCA. From the merged price panel, I extract both monthly and daily versions of the Global Financial Cycle (GFC) proxy using complete return rows rather than zero-filled missing returns.

Figure 1 plots the daily and monthly proxies alongside the GFC factor of [Miranda-Agrippino and Rey \[2020\]](#), using the updated project palette and illustrating their strong co-movement and the ability of the real-time proxy to track global financial conditions reliably.



Source: Author's calculations using a historical input panel (archived Stooq prices plus documented Yahoo backfills) spliced with Yahoo Finance and public exchange APIs. GFC F: Methodology based on Habib & Venditti (2019).
Note: Proxies are the first principal component (PC1) of returns from a panel of global stock indices. The cumulative series are standardized for comparison.
Data available at <https://github.com/iweigandi/daily-global-financial-cycle-proxy>

Figure 1: Daily & Monthly GFC Proxies vs. GFC factor of [Miranda-Agrippino and Rey \[2020\]](#)

In the June 2026 run, the **monthly proxy** explains 45.99% of the total variance in monthly stock market returns. The unsmoothed **daily proxy**, constructed directly from a balanced panel of daily returns, explains 35.35% of the total variance in daily returns. The monthly proxy has a correlation of 0.943 with the Miranda-Agrippino and Rey factor over the overlapping sample. These estimates are based on the price-spliced country/index panel and not on a proxy-level splice.

References

- Aldasoro, I., Hördahl, P., & Zhu, S. (2022). *Under pressure: market conditions and stress*. BIS Quarterly Review, 19, 31–45.
- Davis, J. S., & Zlate, A. (2023). *The global financial cycle and capital flows during the COVID-19 pandemic*. European Economic Review, 156, 104477.
- Habib, M. M., & Venditti, F. (2019). *The global capital flows cycle: structural drivers and transmission channels* (No. 2280). ECB Working Paper.
- Miranda-Agrippino, S., & Rey, H. (2020). *US monetary policy and the global financial cycle*. The Review of Economic Studies, 87(6), 2754–2776.

Implementation Diagnostics

The recovered Stooq archive provides 54 index price series from 2000-01-03 to 2026-06-18. The three index-list entries absent from the archive, Dow Jones Composite, Dow Jones Utilities, and Dow Jones Transportation, are backfilled from Yahoo Finance from 2000-01-03. In the June 2026 update run, replacement sources extended 54 of the 57 input indices; RTS, SOFIX, and SSE B-SHARE remain historical-only until a reliable replacement source is added.

Overlap checks are performed before splicing each replacement price series. The June 2026 run produced 51 pass cases, 3 review cases, and 3 historical-only cases. The review cases are retained in the diagnostics file rather than hidden: MOEX, OMX Stockholm, and OSE. The merged price panel is then used to estimate the daily and monthly PC1 proxies from scratch.